Consent

Thank you for participating in this study. It will include decision-making tasks followed by some questions about yourself.

Estimated study time: 30 minutes.
Your responses are confidential. Whatever information you convey in this study will be anonymized and will not be traced back to you.

Please read all the instructions and information very carefully.
If you have any questions, you may contact Matan Solomon:
ma.solomon@campus.technion.ac.il

HEADS UP
Achieving high performance can qualify you for **bonus** payments of up to **1 pound**.

Notice: After choosing one of the options below, click on the blue arrow at the bottom right side of the screen.

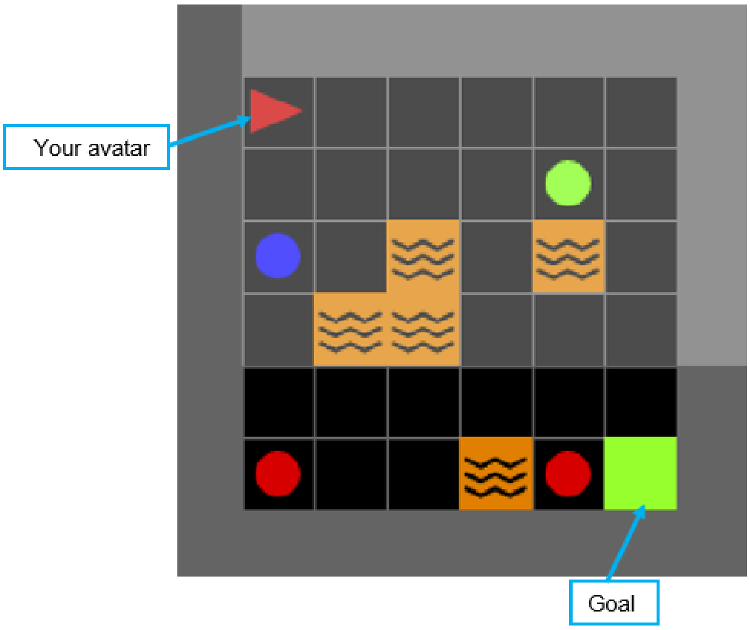
I confirm my participation in the study

I DO NOT confirm my participation in the study

Please record your participant ID in prolific academic:

Game instructions

In this study, you'll play a simple yet strategic game. You control a triangle-shaped avatar in a grid world.
Your objective is to maximize your score by collecting colored balls while minimizing the number of actions you take—every step matters!



Game Rules:
Objective
Guide your avatar to the green goal cell, maximizing points while minimizing actions.

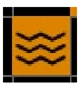
Actions:
The avatar can move forward, turn left or right, or pick up a ball directly in front of it.

Termination
A round ends when either:

1. The avatar reaches the green goal cell.
2. 70 actions have been taken without reaching the goal.

Scoring

- **Balls**
    - Red: -1 points
    - Green: +2 points
    - Blue: +4 points
- **Lava cells**



    - Special hazardous cells, entering (stepping into) a lava cell incurs a -3 point penalty per entry.

- **Action cost:** –0.1 points per action (move, turn, or pick up).

- **Getting to the Goal:** +5 points

For example, if you collect a red ball, you will get: -1.1 points, -1 for the ball, and -0.1 for the pickup action.
When collecting a blue ball, you will get: 4-0.1=3.9 points.

Blocking

- The avatar cannot **move forward** if a **ball** or a **wall** is directly in front of it.
- Attempting to move forward in such a case **has no effect** on position but **still consumes one action** (–0.1 points).

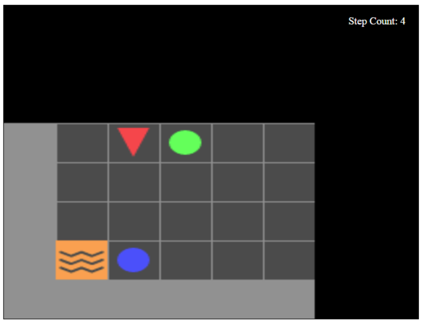
To proceed, the avatar must either:

- **Pick up** the ball (if it's in front) or **navigate around** the obstacle.

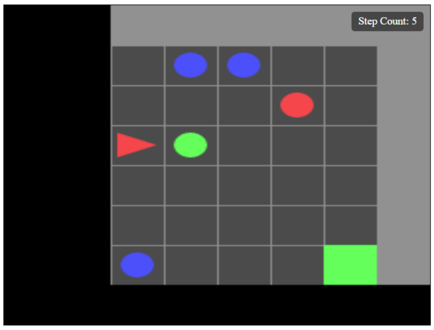
Plan your route to collect high-value balls using as few actions as possible.

Avatar View:

The avatar's view is limited, and it can only see what is in front of it and 3 blocks to each side; out-of-view cells appear darker.
Example images**:**



By turning, the view will also change and let the avatar see different parts of the grid

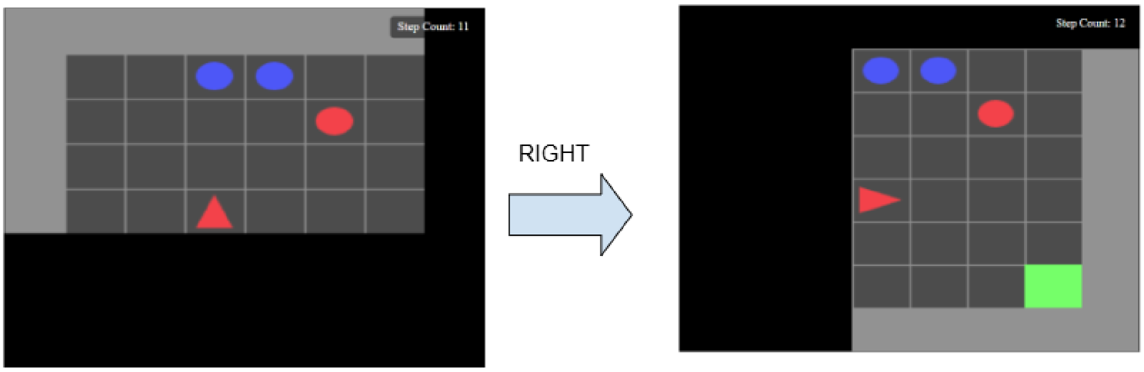


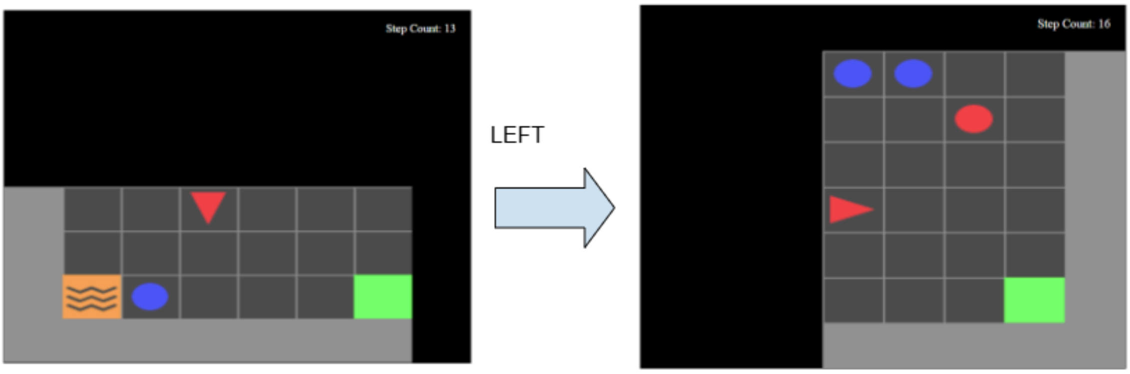
When the avatar is facing a ball, like we see in the upper example, it can collect the ball using the Key ‘1’. There is no limit to the number of balls the avatar can collect.

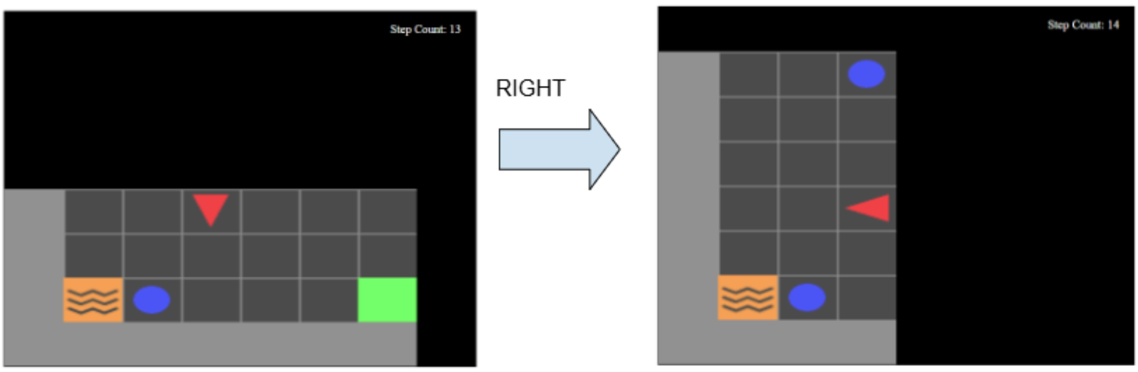
Turning:

The Left and Right turns are from the viewpoint of the Avatar; this is similar to turning your head in some direction, rather than actually taking a turn.

Let's see some examples of turning:



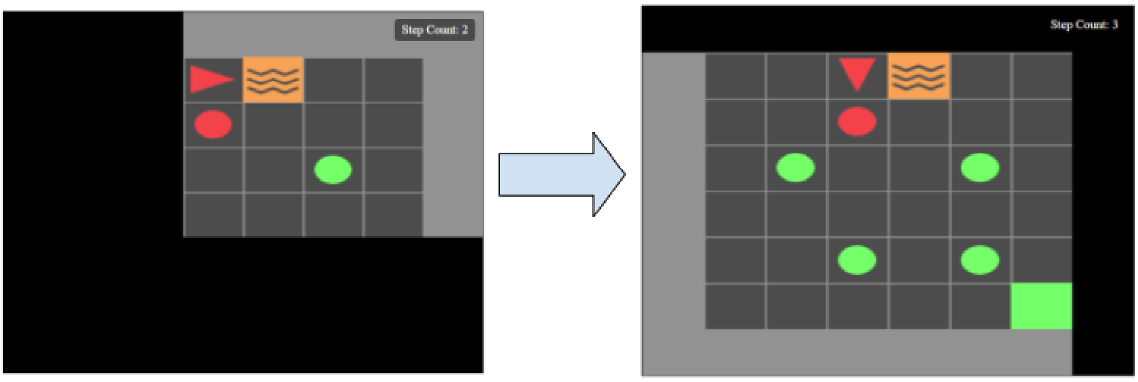




Let's see that you got it

For each pair of images, choose what action was taken and what immediate reward was given.

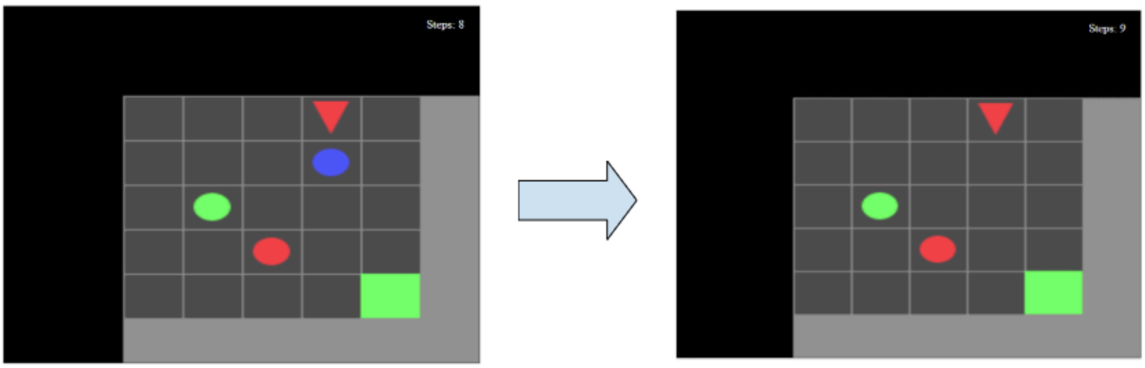
You cannot continue until you answer the question correctly.



- Turn Left
- Turn Right
- Move Forward
- Pickup

Reward?

- 0.1
- -0.1
- 0
- 4

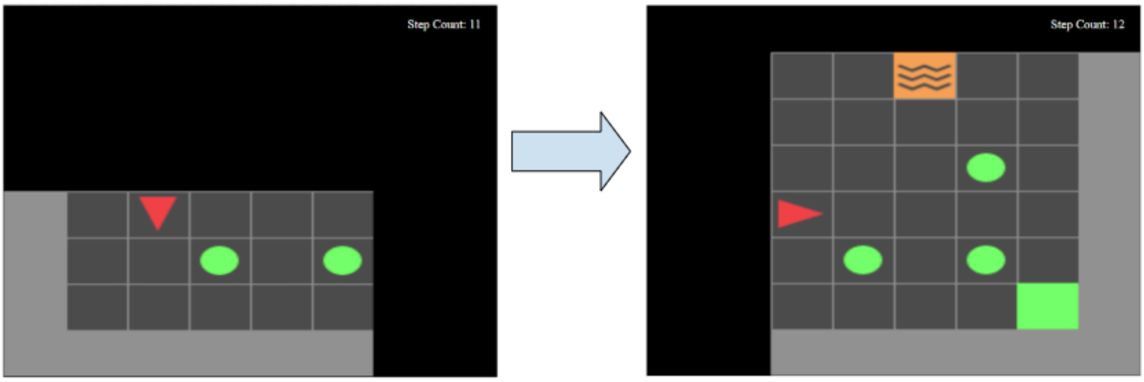


- Turn Left
- Turn Right
- Move Forward

- Pickup

Reward?

- -0.1
- 3.9
- 4
- -4



- Turn Left
- Turn Right
- Move Forward
- Pickup

Reward?

- -0.1
- 0
- 0.1
- 4

Turorial

<u>-------- Practice time - Getting familiar with the Environment --------</u>

In the first part of the game, you will control the avatar using your keyboard:
Up arrow (↑) – Move forward
Left arrow (←) Turn the avatar to ITS left
Right arrow (→) Turn the avatar to ITS right

Key ‘1’ – Pick up a ball directly in front of the avatar

To get a better understanding of the environment and the game, you will play the game yourself. In the link below, you need to play at least 3 rounds of the game.

HEADS UP
After 3 rounds, a green button will appear below. Press it to finish the tutorial and receive the completion number to continue the survey.

Good luck!

Notice: When planning your movement, feel free to **take your time**.

After four rounds, a green button will appear; pressing it will finish the tutorial.

The following link will open the game in a new browser tab - **link**

Paste your confirmation code here:

You are now ready to proceed to the **main task**!

Good luck!

Main Task Control

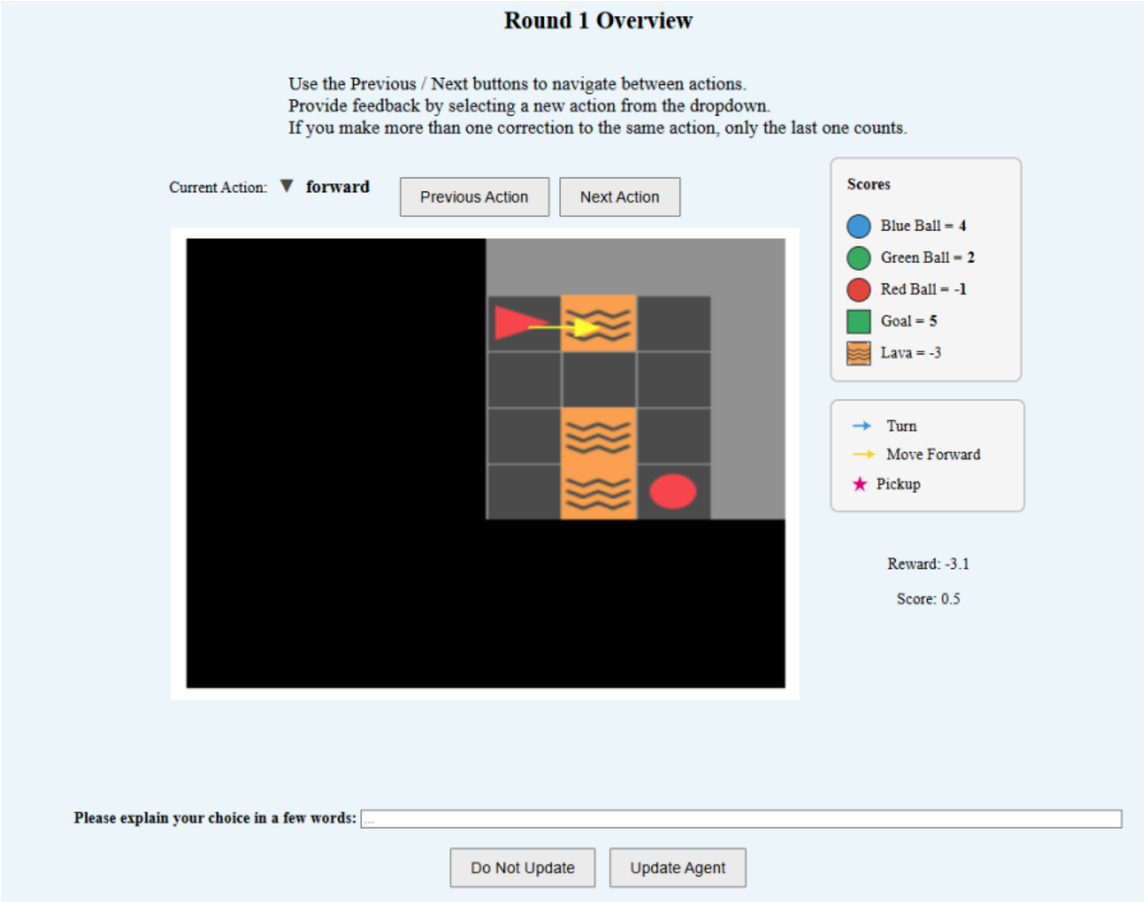
Main Task - AI Agent Assistance

Read the instructions very carefully, and move to the task.

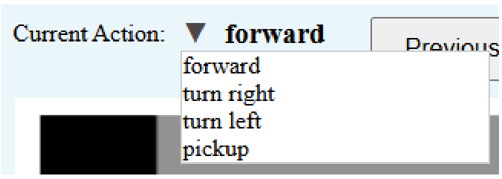
Instructions:

In this section, an AI agent will control the avatar in the game for five rounds.
The agent is imperfect—it may make mistakes or choose suboptimal actions.
Your task is to train the agent to improve its performance: the better it performs by the end of the five rounds, the greater your reward.
In each round, you will first observe the agent controlling the avatar. Afterwards, you will

proceed to the Round Overview Page, where you can provide feedback by correcting the agent's poor decisions.



Use the *Previous* and *Next Action* buttons to step through the agent's most recent path, one action at a time.
When you identify a poor action, click its label (e.g., "forward") and select a better alternative.



If you revise the same action more than once, only your final selection will take effect.

After selecting a correction, it will be highlighted with a blue background, and a white symbol will appear on the avatar to indicate the newly selected action.
In this example, we want to tell the agent not to step into the lava, so we selected the 'turn right' action instead, toward the blue ball.

- When you finish submitting your feedback, click **Update Agent** to help the agent learn from your corrections and adjust its strategy.

If you believe the agent's actions in the round were already optimal, click **Do Not Update** to continue **without** updating the agent.

Main Task

The following link will open the task in a new browser tab - **link**

To continue, enter below the confirmation number you got at the end of the task:

Main Task

Main Task - AI Agent Assistance

Read the instructions very carefully, and after that go to the task

Instructions:

In this section, an AI agent will control the avatar in the game for five rounds.
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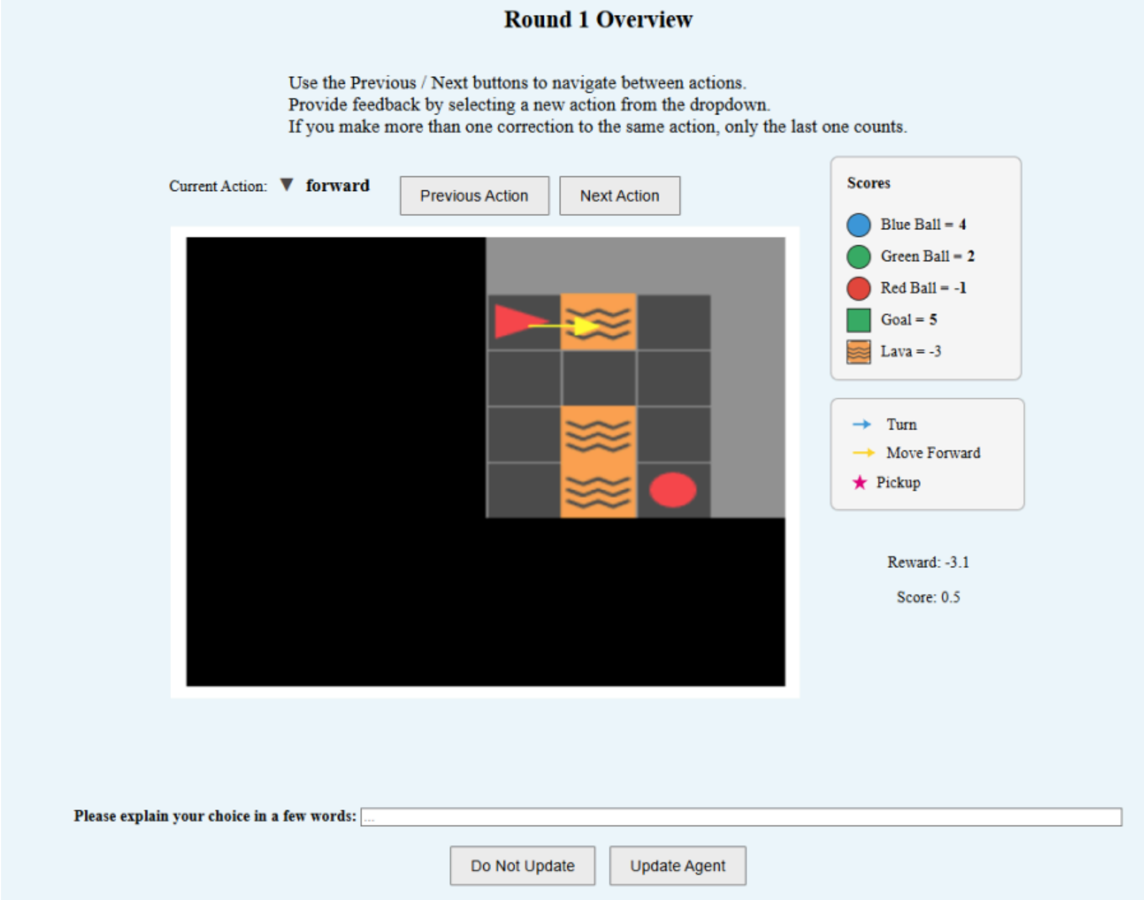

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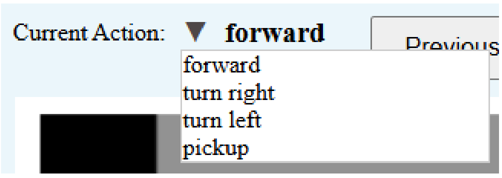

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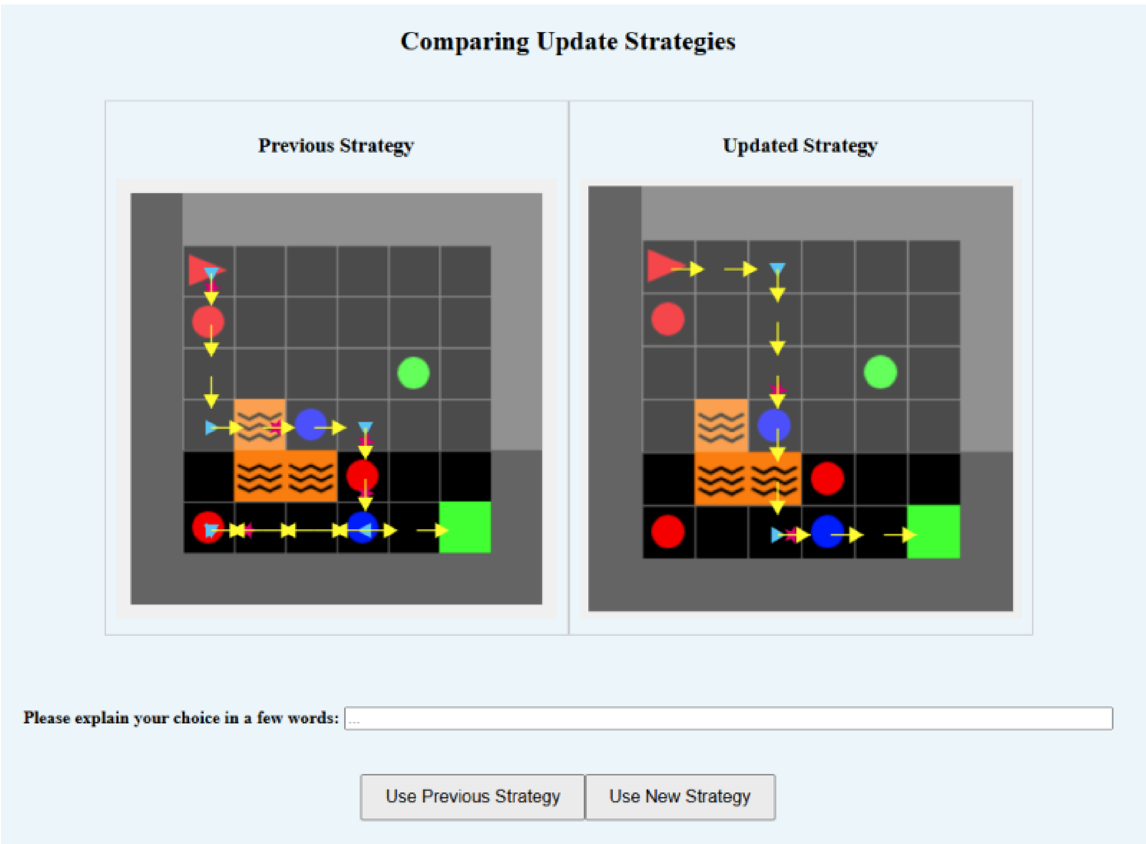
Demonstrating the agent’s new strategy

Agent updates do not always produce the expected outcome; sometimes new, unintended behaviors emerge.
To help with this, we present the previous and updated agents side by side. After watching them perform under the same conditions, you can simply choose which version you would like to continue with.

Below is an example comparing an agent’s strategy before and after feedback.

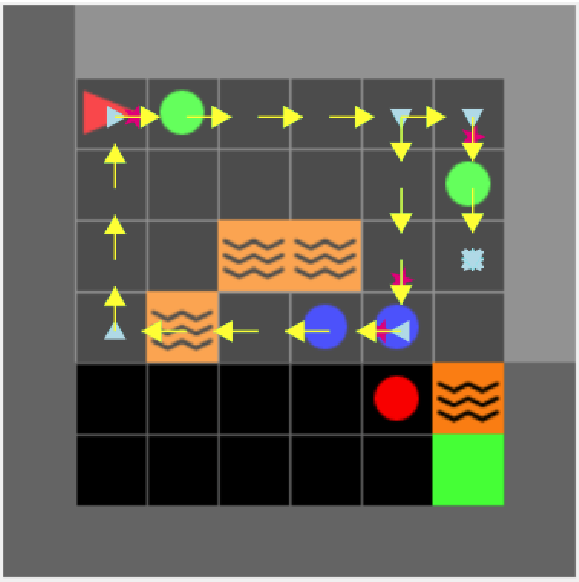
- In the **previous strategy**, the avatar steps into a lava cell (-3) and collects red balls (-1).
- In the **updated strategy**, the avatar avoids collecting red balls but still steps into the lava —making it imperfect, yet better than the earlier version.



In the comparison display, each icon represents a specific action:

- **Yellow Arrow:** - **moving forward** one step.
- **Light-blue Arrow** - **turn** to the indicated direction.
- **Pink Star** - **picking up** the ball directly ahead.

As noted earlier, the AI agent's behavior can occasionally appear irrational.
In certain states, it may fail to select an optimal action, execute illegal actions like moving into a wall, or repeat turning actions without progress, effectively causing the avatar to become "stuck" (see example below).
In a path demonstration, such behavior is visible as follows:



Main Task

The following link will open the game in a new browser tab - link

to continue, enter below the confirmation number you got at the end of the task:

System Satisfaction

Answer the following questions based on your opinion of the system.

	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree
The explanations of the agent's paths were clear and easy to understand	○	○	○	○

	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree
This is an attention check question, please select 'Strongly agree'	○	○	○	○
The explanations felt too detailed and overwhelming	○	○	○	○
The explanations helped me see how my feedback changed the agent's behavior	○	○	○	○
The visual demonstrations of policy changes helped understand the differences between agents	○	○	○	○

Part 3: Trust the Agent

Evaluate the agent

	Much worse	Worse	Somewhat worst	Same	Somewhat better	Better	Much better
How would you evaluate the agent's performance relative to yours in the game?	○	○	○	○	○	○	○

Final Part
The bonus for this survey will be proportional to the success of the avatar in the next game round.
Would you like to let the trained agent (after completing the five training rounds) control the avatar and finish the survey,
or would you prefer to control the avatar yourself?

- ○ I will play myself
- ○ I will let the AI agent play

Please briefly explain your choice:

Part3 - Play Yourself

The following link will open the game in a new browser tab - **link**
Paste your confirmation code here:

Demographics

Finally, some background details:

Age:

Gender identity:

Male Female

Non-Binary

Prefer not to say

In which country do you reside?

- United states
- United Kingdom
- Australia
- Canada
- Other

Years of education:

- Less than 8 years
- 8-12 years
- Bachelor level academic education
- Master level academic education
- PhD, MD, or other doctoral degree
- Other:

How familiar are you with Artificial Intelligence?

- ○ Not at all familiar
- ○ Slightly familiar (basic awareness, e.g., from news or media)
- ○ Somewhat familiar (some knowledge, e.g., from a course or self-study)
- ○ Moderately familiar (practical experience, e.g., using AI-powered tools or assistants)
- ○ Very familiar (frequent user or hobbyist, e.g., building AI models or applications)
- ○ Extremely familiar (professional experience, e.g., working in an AI-related field or industry)
- ○ Expert level (advanced expertise, e.g., conducting research, developing new AI techniques or technologies)

Please state your agreement with the following sentences **regarding Artificial Intelligence (AI)**:

	1. Strongly disagree	2	3	4. Neither agree nor disagree	5	6	7. Strongly agree
I believe AI will improve my life.	○	○	○	○	○	○	○
I believe that AI will improve my work.	○	○	○	○	○	○	○
I think I will use AI technology in the future.	○	○	○	○	○	○	○
I think AI technology is a threat to humans.	○	○	○	○	○	○	○
I think AI technology is positive for humanity.	○	○	○	○	○	○	○

Summary

Thank you for your participation.

Please share any thoughts or comments about this study in the space below.

Click the next button to complete the survey.

External links:

Tutorial link - https://dpu-tutorial-app.yellowmushroom-27e70244.westeurope.azurecontainerapps.io

Main Task link - https://dpu-main-app.yellowmushroom-27e70244.westeurope.azurecontainerapps.io/?group=2

Final Part link - https://dpu-final-app.salmonwave-632f84f7.eastus.azurecontainerapps.io